This is the Name of my Thesis

by

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This is the Name of my Thesis

Thesis directed by Prof. Ed Visor

Often the abstract will be long enough to require more than one page, in which case the macro "\OnePageChapter" should *not* be used.

But this one isn't, so it should.

Dedication

To all of the fluffy kitties.

Acknowledgements

Here's where you acknowledge folks who helped. But keep it short, i.e., no more than one page, as required by the Grad School Specifications.

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Chapter 1

Introduction

This sample document illustrates how to use the `thesis` class, originally written by John P. Weiss. Some requirements of the Graduate School are written into that file; page size, line spacing, appropriate placement of captions for tables and figures, etc. Revisions by Hongcheng Ni make it possible to use the (optional) `\usepackage{hyperref}` command to enable internal hyperlinks in the final PDF document. Other tasks of conforming to the requirements are left to other existing LaTeX packages. For example, a common problem is to insert graphics — figures and tables — into the body of the thesis. For this one should use the `graphicx` package, which is part of the standard TeX distribution. Likewise, the Grad School specs say that a large table may be displayed in landscape mode at reduced size, but its caption must also be in rotated position, in the same font and size as the normal text in the body of the thesis. To accomplish this, the user must invoke the `rotating` package, available online.

Figure 1.1 shows an image from a PDF file imported into this document using the `graphicx` package. The command `\usepackage{graphicx}`, which appears near the very top of the main LaTeX file, reads in this package which defines the `\includegraphics{}` macro.

1.1 Lists in `thesis` class

In `thesis` class (for Colorado University), lists are defined so that nested lists will be numbered or marked appropriately. First, an itemized (non-enumerated) list prefaces each item with a bullet. Nested itemized list use asterisks, then dashes, then dots. These lists are typed between

Figure 1.1: This diagram of a cylinder and various measurements and quantities was actually made using **xfig**, a freeware drawing program for Unix systems. Diagrams can be exported directly to PDF files, the preferred format for vector graphics. Vector graphics can be magnified indefinitely without degradation, whereas bitmap images (JPG and PNG) must be pretty high-resolution if you don't want them looking all pixellated when magnified.

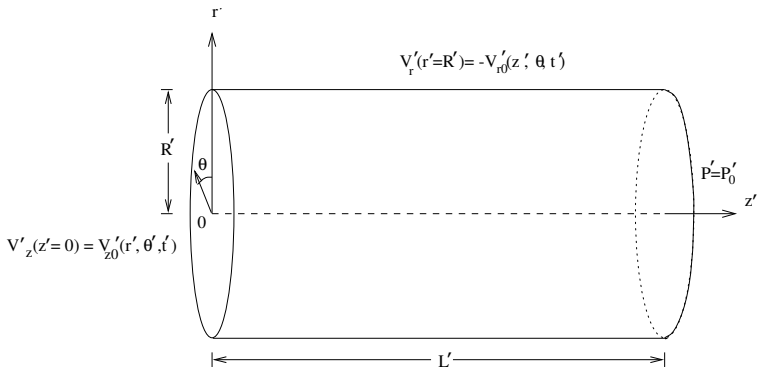
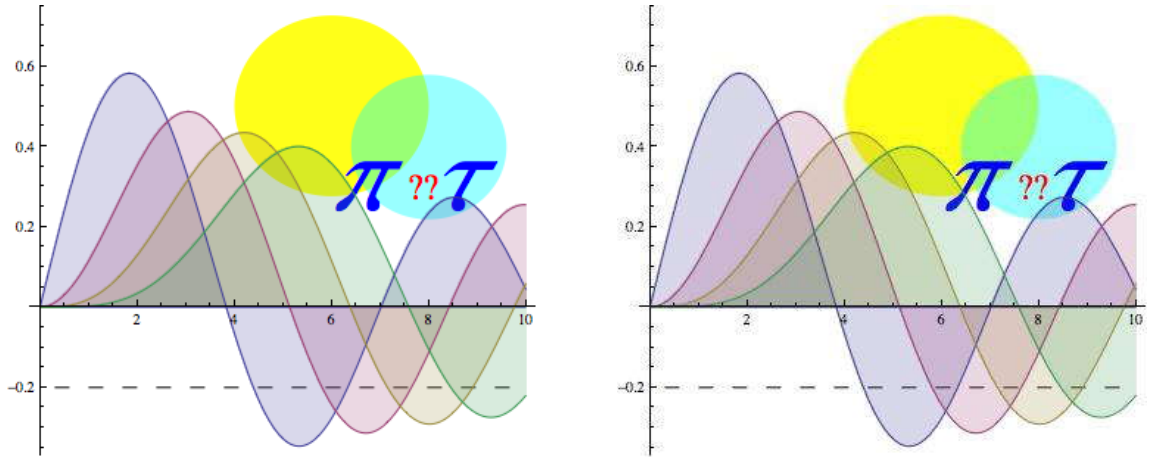


Figure 1.2: The JPEG bitmap format is great for photos but crummy for diagrams (including drawings, graphs, charts) because it can't gracefully handle sharp edges. Note the same bitmap image below from a PNG file and from a JPG file; the latter shows characteristic "ringing" at sharp edges – including text! Seriously, magnify and look closely at the JPG's awful lines and edges. Vector-format PDF is the best for diagrams, but if you must use a bitmap image, let it be PNG. (Left: file *drawing.png*. Right: file *drawing.jpg*.)



the `\begin{itemize}` and `\end{itemize}` commands.

- This is “itemized” item A.
- This is “itemized” item B.
- This is “itemized” item C.
  - This is “itemized” subitem A.
    - This is “itemized” subsubitem A.
      - This is “itemized” subsubsubitem A.
    - This is “itemized” subsubitem B.
  - This is “itemized” subitem B.
- This is “itemized” item D.

Enumerated lists use the commands `\begin{enumerate}` and `\end{enumerate}`, and nested enumerations appear like this.

(1) This is “enumerated” item A.

(2) This is “enumerated” item B.

(3) This is “enumerated” item C.

  (a) This is “enumerated” subitem A.

    (i) This is “enumerated” subsubitem A.

      (i.a) This is “enumerated” subsubsubitem A.

    (ii) This is “enumerated” subsubitem B.

  (b) This is “enumerated” subitem B.

(4) This is “enumerated” item D.

The work presented here[1] is an extension of Lao[7] and Lao et al.[8], fictional references that are in the bibliographic source file `refs.bib`.

Table 1.1: Here is an example of a table with its own footnotes. Don't use the `\footnote` macro if you don't want the footnotes at the bottom of the page. Also, note that in a thesis the caption goes **above** a table, unlike figures.

wave form	S (kVA)	P (kW)	Q^* (kVAr)	$D^\dagger$ (kVAd)
Fig. 1.1a	25.48	25.00	-2.82	4.03
Fig. 1.1b	25.11	18.02	-9.75	14.52
Table 2.1	24.98	22.26	9.19	6.64
Table 1.1	23.48	15.00	6.59	16.82
Fig. 2.1	24.64	22.81	-0.44	9.3

*kVAr means reactive power.
†kVAd means distortion power.

[1] Footnotes are handled neatly by LaTeX.

Chapter 2

Mathematical Formulation

The objective of this fake thesis document is to demonstrate a multitude of LaTeX features as well as features specific to the thesis class. We start by giving one short formula, and one big hairy multi-line formula (one of the non-dimensional Navier-Stokes equations):

$$A = \pi r^2 \tag{2.1}$$

$$\begin{aligned}
\rho\left[\frac{DV_r}{Dt} - M\epsilon^2\frac{V_\theta^2}{r}\right] &= -\frac{\delta^2}{\gamma\ M}\frac{\partial P}{\partial r} + \frac{M\ \delta^2}{Re}\left\{2\frac{\partial}{\partial r}\left[\mu\left(\frac{\partial V_r}{\partial r} - \frac{1}{3}\nabla\cdot\overline{\mathbf{V}}\right)\right]\right. \\
&\quad + \frac{1}{r}\frac{\partial}{\partial\theta}\left[\mu\left(\frac{1}{r}\frac{\partial V_r}{\partial\theta} + \epsilon\frac{\partial V_\theta}{\partial r} - \epsilon\frac{V_\theta}{r}\right)\right] \\
&\quad + \frac{\partial}{\partial z}\left[\mu\left(\frac{1}{\delta^2}\frac{\partial V_r}{\partial z} + \frac{\partial V_z}{\partial r}\right)\right] \\
&\quad \left. + 2\ \frac{\mu}{r}\left[\frac{\partial V_r}{\partial r} - \frac{\epsilon}{r}\frac{\partial V_\theta}{\partial\theta} - \frac{V_r}{r}\right]\right\},
\end{aligned} \tag{2.2}$$

2.1 Explanation of equations

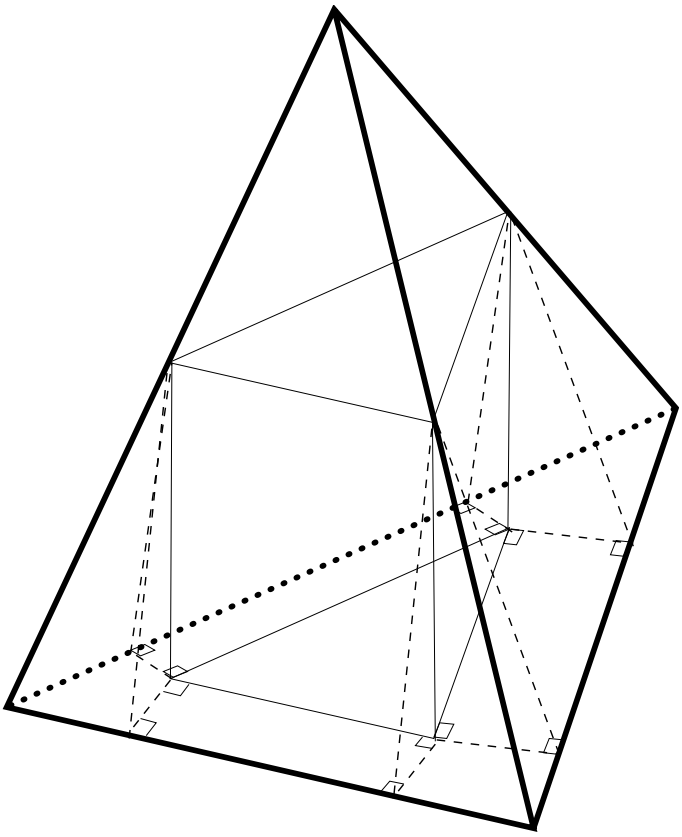
The latter equation is non-dimensionalized using the following definitions:

$$r = \frac{r'}{R'}, \quad z = \frac{z'}{L'}, \quad t = \frac{t'}{t'_a}, \quad \kappa = \frac{\kappa'}{\kappa'_0}, \quad \mu = \frac{\mu'}{\mu'_0}, \quad C_V = \frac{C'_V}{C'_{V0}},$$

where P'_0 is the initial static pressure in the cylinder, and ρ'_0 and T'_0 are the density and temperature of the fluid being injected from the sidewall.

Here is an example of using the macros `\singlespacing` and `\doublespacing`:

Figure 2.1: A triangular pyramid may be cut up as shown, to yield one top pyramid (with one-eighth the volume of the full pyramid), three bottom corner pyramids (which, when joined, are congruent to the top pyramid), three prisms along the bottom edges (the area of whose bottom faces total $B/2$) and the large central prism (volume $= (B/4)(h/2) = Bh/8$). The image, from PDF file "pyr.pdf", was read in using the `\includegraphics` command, from the `graphicx` package.



This paragraph was preceded by the command `\singlespacing`. See the Specifications of the Grad School for instructions about when single spacing is appropriate in a thesis.

And now, here is an example of using the macros `\begin{singlespace}` and `\end{singlespace}`; another way to get single-spacing.

Two cases are studied in the present work which differ only in the boundary conditions. Each different boundary condition model a different source of instability. The boundary of the first case consists of a steady, axisymmetric sidewall radial velocity boundary and a time-dependent, non-axisymmetric endwall axial velocity boundary. The second case is studied with a fixed impermeable axial velocity along the endwall and a combination axisymmetric steady and non-axisymmetric unsteady radial velocity along the sidewall.

Usually you want to use a table produced by some other software, such as Excel, rather than try to do it using LaTeXmacros. If the table is saved/printed to a PDF file, then it can be displayed using the `\includegraphics` macro inside a `table` environment:

Some of the boundary conditions are:

$$z = 0; \qquad V_z = \begin{cases} 0, & t \le 0 \\ \\ \widetilde{F}_{zw}(r,\theta,t), & t > 0 \end{cases} \tag{2.3}$$

$$z = 0; \qquad V_\theta = V_r = 0 \tag{2.4}$$

$$r = 0; \qquad P, \rho, T, V_r, V_\theta, V_z \text{ finite}, \tag{2.5}$$

$$r = 1; \qquad V_r = F_{rws}(z), \tag{2.6}$$

$$r = 1; \qquad V_z = V_\theta = 0, \tag{2.7}$$

and solutions must be periodic in θ.

If you don't believe this stuff, check out Mulick[10] and Baylor[1].

2.2 Yet another section

2.2.1 Just meaningless text to test lines per page

According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document

Table 2.1: This table wasn't constructed with LaTeX commands, but resides in PDF file (`tableD.pdf`) created by some other software.

n	n^2	n^3	n^4	n^7	n^{13}
2	4	8	16	128	8192
3	9	27	81	2187	1594323
4	16	64	256	16384	67108864
5	25	125	625	78125	1220703125
6	36	216	1296	279936	13060694016
7	49	343	2401	823543	96889010407

conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines of print per page of a thesis. This should be true whether the font size is 10, 11, or 12. Count them up; does this document conform? According to the Grad School specs. there should be 24–27 lines

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What is it? This is a labelled paragraph. The heading of the paragraph is emphasized. This is a labelled paragraph. The heading of the paragraph is emphasized.

2.2.2 This is a subsection

This is a subsection. Filler filler filler filler filler filler filler filler. Filler filler filler filler filler filler filler filler.

2.2.3 This is another subsection

This is another subsection. Filler filler filler filler filler filler filler filler. Filler filler filler filler filler filler filler filler.

This is paragraph number 2. It used a `\paragraph{}` header, which are always inlined (with extra space) and boldfaced.

This is the third paragraph of the subsection. Filler filler filler filler filler filler filler filler. Filler filler filler filler filler filler filler filler.

2.2.3.1 This is a subsubsection (1)

This is the first paragraph of the subsubsection. Whether it is numbered or inlined depends on the option selected at the beginning of the thesis.

By default, a `\subsubsection` heading is numbered and set off on a separate line, left-justified.

However. Using the `inlineh4` option, subsubsection headers are inlined. And using the

`nonumh4` option suppresses numbering of the subsubsections. Together they make subsubsection headings just the same as paragraph headings.

2.2.3.2 This is another subsubsection (2)

Once again, whether its heading is numbered and/or inlined depends on the class options chosen at the start.

There is no "subsubsubsection" entity, and "subparagraph" gets no special treatment in **thesis** class.

2.3 The End

Finally, this is the end. The bibliography starts on the next page. Note how the `\hyperref` package (mentioned in chapter 1) also makes hyperlinks from references (e.g., Mulick[10]) to entries in the bibliography.

Bibliography

[1] G. I. Baylor. Up, up and away. Proc. Roy. Soc., London A, 294:456–475, 1959.

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Appendix A

Notes on the Sample Data

About appendices: Each appendix follows the same page-numbering rules as a regular chapter; the first page of a (multi-page) appendix is not numbered. Tables and figures in an appendix are numbered by its letter, as Table A.1 shows. The plots, measurements and notes below are invented for this example and describe no real study.

Table A.1: Invented measurements for six sample plots, included only to show how a table is numbered and captioned inside an appendix.

plot	area (m^2)	rows	seedlings	survival (%)
A	12.0	4	96	91
B	10.5	3	72	88
C	14.2	5	120	94
D	9.8	3	66	79
E	11.6	4	90	86
F	13.1	4	104	92

The numbered notes that follow record how the invented data in Table A.1 were gathered and coded. A list like this is a common use for an appendix: it keeps procedural detail out of the main chapters while leaving it available to a careful reader.

(1) Each plot was marked out with four corner stakes and a length of string, and its area was measured twice. The table reports the mean of the two measurements, rounded to one decimal place.

(2) Rows were laid out along the longer side of each plot. Where a plot was too narrow for a full row, the partial row was left unplanted and is not counted.

(3) Seedlings were planted at a fixed spacing within each row. The seedling count is the number planted on the first day, not the number present at the end of the season.

(4) Survival is the share of planted seedlings still alive at the final visit, expressed as a whole-number percentage.

(5) Visits took place once a week. A visit that was missed because of weather was made up on the next dry day and recorded under its original week.

(6) A seedling was counted as alive if it had at least one green leaf. Seedlings that had been uprooted but were replanted on the same visit were counted as alive.

(7) Plot D lies closest to the path and lost several seedlings to foot traffic early in the season. No adjustment was made for this in the table.

(8) Watering followed the same schedule in every plot. On the two hottest weeks an extra watering was added to all six plots at once, so the schedule stayed the same across plots.

(9) No fertilizer was applied. A single layer of mulch was spread over every plot after the third visit.

(10) Weeds were removed by hand at each visit. The time spent weeding was noted but is not reported here.

(11) Measurements were written in a field notebook and typed into a spreadsheet on the same evening. Each typed entry was checked against the notebook by a second reader.

(12) Two entries were corrected during that check: one transposed digit in the area of Plot B and one miscounted row in Plot E. The table shows the corrected values.

(13) All numbers in this appendix are fictional. They were chosen to fill a table of realistic size, and no conclusion should be drawn from them.

(14) Readers adapting this example for their own thesis should replace this appendix entirely with their own material, keeping only the structure if it is useful.

Appendix B

A Verse on Layout

(Written for this example) (A one-page chapter — page must be numbered!) This appendix shows how a short, verse-like block sets inside a one-page chapter. The lines below were written for this sample document and quote no other work; their subject is the layout that the class itself provides.

> The page begins with level margins, quiet, wide and white,
> a frame that holds the argument and keeps the columns right.
> The chapters open, numbered, on a fresh and empty sheet;
> the sections follow, step by step, in order clean and neat.
> Each figure waits beside its words, its caption placed below;
> each table wears its caption high, the first thing it will show.
> The footnotes speak in smaller type beneath the running prose,
> and citations send the reader back to where each claim arose.
> The contents list the road ahead, the pages mark the way,
> and appendices hold the rest the chapters could not say.
> When every line is in its place and every number true,
> the document is finished, and it waits for someone new.

(A short prose note between the two blocks shows how the class returns to body text after a quotation and then sets a second one.)

> So here's to every rule that keeps the layout in its place:
> the spacing and the numbering, the headings and the face.
> You never read a single line, nor weigh what has been planned,
> yet every thesis leans on you, a steady, silent hand.